# Binary & Ternary Search

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# Binary Search

- So far we only knew one way to look up an item - Linear Search.
- When the given data is sorted, it lets us search items faster.

# Membership problem

- Given n integers, Q queries are asked. In each query you are given x, print whether x is in the array or not.

# Lowerbound problem

- Given n integers, Q queries are asked. In each query you are given x.
- Print the smallest number a[i] >= x.

# A fit for all situation binary search template

```
int lo = 0, hi = n - 1, ans = -1;
while (lo <= hi) {
    int mid = (lo + hi) / 2;

    if (a[mid] >= x) {
        // write your decisions here
    } else {
        // write your decisions here
    }
}
```

# Upperbound problem

- Given n integers, Q queries are asked. In each query you are given x.
- Print the smallest number a[i] > x.

# Builtin Functions

```
vector<int> v;
// suppose v is a sorted array

auto iter = lower_bound(v.begin(), v.end(), x);

int l_idx = lower_bound(v.begin(), v.end(), x) - v.begin();
```

https://codeforces.com/edu/course/2/lesson/6/1/practice/contest/283911/problem/D

You are given an array $a$ of $n$ integers $a_1, a_2, \ldots, a_n$.

Your task is to response to the queries like "`How many numbers' values are between` $l$ `and` $r$?".

## Input

The first line of the input contains $n$ — the length of the array ($1 \le n \le 10^5$).

The second line contains $n$ integers $a_1, a_2, \ldots, a_n$ ($-10^9 \le a_i \le 10^9$).

The third line contains integer $k$ — the number of queries ($1 \le k \le 10^5$).

The following $k$ lines contain a pair of integers $l$ $r$ — query, described above ($-10^9 \le l \le r \le 10^9$).

## Output

The output must consist of $k$ integers — responses for the queries.

https://atcoder.jp/contests/abc248/tasks/abc248_d

You are given a sequence of length $N$: $A = (A_1, \ldots, A_N)$.

Answer $Q$ queries given in the following format.

- You are given integers $L$, $R$, and $X$. Find the number of elements among $A_L, \ldots, A_R$ whose values are equal to $X$.

## Constraints

- $1 \leq N \leq 2 \times 10^5$
- $1 \leq A_i \leq N$
- $1 \leq Q \leq 2 \times 10^5$
- $1 \leq L \leq R \leq N, 1 \leq X \leq N$, for each query.
- All values in input are integers.

How to solve this problem if a[i] <= 10^9?

https://atcoder.jp/contests/abc186/tasks/abc186_d

Given are $N$ integers $A_1, \ldots, A_N$.

Find the sum of $|A_i - A_j|$ over all pairs $i, j$ such that $1 \leq i < j \leq N$.

In other words, find $\sum_{i=1}^{N-1} \sum_{j=i+1}^{N} |A_i - A_j|$.

## Constraints

- $2 \leq N \leq 2 \times 10^5$
- $|A_i| \leq 10^8$
- $A_i$ is an integer.

How to solve this problem if we had two arrays, and required |A[i] - B[j]| ?

# The big picture of binary search

- So far it seems you need a sorted array to do things like upper/lower bound tasks on them.
- But binary search can do much more than that.
- What does binary search actually find?
    - It finds a changing point of a bitonic function

# Binary Search on the Answer

- Suppose a certain property did not exist, but at some point in time it started existing.
  - Example: The world with pyramids in it could be a property.
- Now, you can ask me about any year and I can tell you whether that property (pyramids being built or not) hold true.
- Notice, you can binary search to find out when the pyramids were built.
- This is known as Binary searching on the answer (to the question - when was the pyramids built?)

https://codeforces.com/contest/702/problem/C

# C. Cellular Network

time limit per test: 3 seconds
memory limit per test: 256 megabytes

You are given $n$ points on the straight line — the positions ($x$-coordinates) of the cities and $m$ points on the same line — the positions ($x$-coordinates) of the cellular towers. All towers work in the same way — they provide cellular network for all cities, which are located at the distance which is no more than $r$ from this tower.

Your task is to find minimal $r$ that each city has been provided by cellular network, i.e. for each city there is at least one cellular tower at the distance which is no more than $r$.

If $r = 0$ then a tower provides cellular network only for the point where it is located. One tower can provide cellular network for any number of cities, but all these cities must be at the distance which is no more than $r$ from this tower.

## Input

The first line contains two positive integers $n$ and $m$ ($1 \le n, m \le 10^5$) — the number of cities and the number of cellular towers.

The second line contains a sequence of $n$ integers $a_1, a_2, ..., a_n$ ($-10^9 \le a_i \le 10^9$) — the coordinates of cities. It is allowed that there are any number of cities in the same point. All coordinates $a_i$ are given in non-decreasing order.

The third line contains a sequence of $m$ integers $b_1, b_2, ..., b_m$ ($-10^9 \le b_j \le 10^9$) — the coordinates of cellular towers. It is allowed that there are any number of towers in the same point. All coordinates $b_j$ are given in non-decreasing order.

## Output

Print minimal $r$ so that each city will be covered by cellular network.

# Binary Search on the Answer Notes

- You should try to fix one/few parameters in your problems.
- You can always do linear search on that parameter.
- But, if that parameter needs to satisfy a property, which:
  - If doesn’t occur at some point, never occurs before it, and
  - If occurs at some point, stays in the future as well
- Then you can do binary search along that parameter and improve performance drastically.

https://cses.fi/problemset/task/1085

You are given an array containing $n$ positive integers.

Your task is to divide the array into $k$ subarrays so that the maximum sum in a subarray is as small as possible.

**Input**

The first input line contains two integers $n$ and $k$: the size of the array and the number of subarrays in the division.

The next line contains $n$ integers $x_1, x_2, \ldots, x_n$: the contents of the array.

**Output**

Print one integer: the maximum sum in a subarray in the optimal division.

**Constraints**

- $1 \le n \le 2 \cdot 10^5$
- $1 \le k \le n$
- $1 \le x_i \le 10^9$

**Example**

Input:
```
5 3
2 4 7 3 5
```

Output:
```
8
```

Explanation: An optimal division is $[2,4], [7], [3,5]$ where the sums of the subarrays are $6, 7, 8$. The largest sum is the last sum $8$.

https://codeforces.com/edu/course/2/lesson/6/2/practice/contest/283932/problem/D

# D. Children Holiday

time limit per test: 2 seconds

memory limit per test: 512 megabytes

The organizers of the children's holiday are planning to inflate $m$ balloons for it. They invited $n$ assistants, the $i$-th assistant inflates a balloon in $t_i$ minutes, but every time after $z_i$ balloons are inflated he gets tired and rests for $y_i$ minutes. Now the organizers of the holiday want to know after what time all the balloons will be inflated with the most optimal work of the assistants, and how many balloons each of them will inflate. (If the assistant has inflated the balloon and needs to rest, but he will not have to inflate more balloons, then it is considered that he finished the work immediately after the end of the last balloon inflation, and not after the rest).

## Input

The first line of the input contains integers $m$ and $n$ $(0 \le m \le 15000, 1 \le n \le 1000)$. The next $n$ lines contain three integers each, $t_i$, $z_i$, and $y_i$, respectively $(1 \le t_i, y_i \le 100, 1 \le z_i \le 1000)$.

## Output

In the first line print the number $T$, the time it takes for all the balloons to be inflated. On the second line print $n$ numbers, the number of balloons inflated by each of the invited assistants. If there are several optimal answers, output any of them.

# Sqrt(x) problem

- You are given x <= 10^18. Find sqrt(x).
- You can’t use the built in function.

https://codeforces.com/edu/course/2/lesson/6/2/practice/contest/283932/problem/E

# E. Equation

time limit per test: 2 seconds

memory limit per test: 256 megabytes

Find a number $x$ such that $x^2 + \sqrt{x} = c$.

## Input

The input contains one real number $C$ ($1.0 \le c \le 10^{10}$).

## Output

Print one number, the required $x$. The answer will be considered correct if the relative or absolute error is not more than $10^{-6}$.

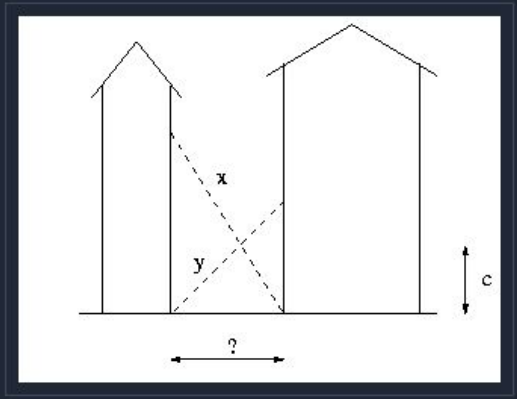
## Examples

| input |
| --- |
| 2.0 |
| **output** |
| 1.0 |

| input |
| --- |
| 15.6 |
| **output** |
| 3.698232168829691 |

https://lightoj.com/problem/crossed-ladders

A narrow street is lined with tall buildings. An **x** foot long ladder is rested at the base of the building on the right side of the street and leans on the building on the left side. A **y** foot long ladder is rested at the base of the building on the left side of the street and leans on the building on the right side. The point where the two ladders cross is exactly **c** feet from the ground. How wide is the street?




## Input

Input starts with an integer **T (≤ 10)**, denoting the number of test cases.

Each test case contains three positive floating point numbers giving the values of **x**, **y**, and **c**.

## Output

For each case, output the case number and the width of the street in feet. Errors less than $10^{-6}$ will be ignored.

# Ternary Search

- Instead of checking with one midpoint, we evaluate the function at two thirds, and then make our decision.
- A good question to ask is what this gives us?
- Many people teach this wrong and say that this is an improved version of the binary search and both do the same thing. But it’s not like that.
- The ternary search is used to find the highest/lowest point of a concave/convex function.

# Peak/Valley of a parabola

- Given A (> 0), B, C what is the value of x for which the following function has the lowest value?

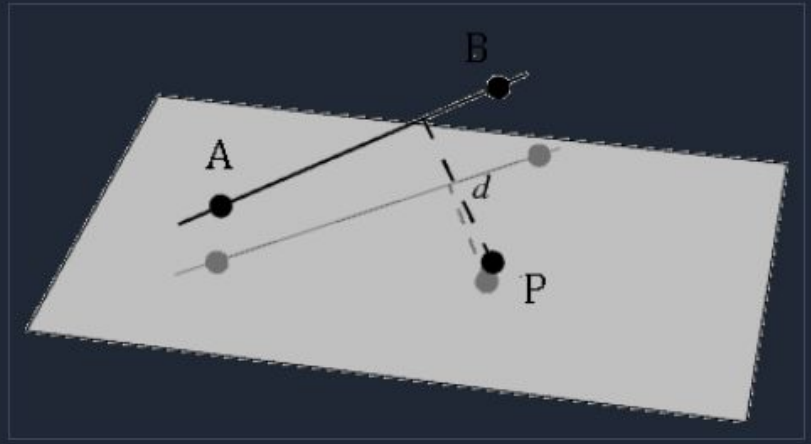
$$f(x) = Ax^2 + Bx + C$$

# A warning on discrete functions

- Sometimes the function might be discrete (e.g. defined only on integers).
- In that case, the function MUST satisfy unimodality
  - Unimodal: When the function is strictly increasing and then strictly decreasing (or vice versa). In other words, the discrete values must form a single peak or a single valley without multiple local extrema.
- The derivative (finite difference) trick: Instead of ternary searching over f(x), you can consider binary searching on f(x) - f(x - 1). The sign of this value only changes at the peak/valley.

https://lightoj.com/problem/point-segment-distance-3d

Given a segment in a 3D space, identified by **A($x_1$, $y_1$, $z_1$)**, **B($x_2$, $y_2$, $z_2$)** and another point **P(x, y, z)** your task is to find the minimum possible Euclidean distance between the point **P** and the segment **AB**.




## Input

Input starts with an integer **T (≤ 10000)**, denoting the number of test cases.

Each case starts with a line containing nine integers **$x_1$, $y_1$, $z_1$, $x_2$, $y_2$, $z_2$, x, y, z**. The magnitude of any integer will not be greater than **100**.

## Output

For each case, print the case number and the distance. Errors less than $10^{-6}$ will be ignored.

# Egg Dropping Puzzle

- You are currently on a 100 storied building and have TWO eggs with you.
- You can throw an egg from a floor and verify whether the egg breaks.
- Both eggs have the same strength. And if the egg doesn't break after falling, assume that it doesn't cause any internal damage (so it's not weakened in future).
- You want to find the highest floor, from where the egg doesn't break when thrown.
- Find a strategy to minimize the number of throws required. You can break both the eggs in the process if needed.